המחלקה למדעי המחשב במכללה האקדמית ספיר

# מתכנן יום חכם | Smart Day Planner

מערכת AI שמארגנת מחדש את היום לפי מצב המשתמש, אנרגיה, משימות ומשוב בזמן אמת.


מגישות: אמינה פראחין וסארה אלחמידי
מנחה: ד"ר האופטמן עמי


React + Node/Express + AI Modules

מדעי המחשב
המכללה האקדמית ספיר (ע"ר)

AI **מערכת AI לתכנון יום דינמי לפי מצב המשתמש** המערכת מתאימה את הלו״ז בזמן אמת לפי מצב המשתמש והמשימות שלו.

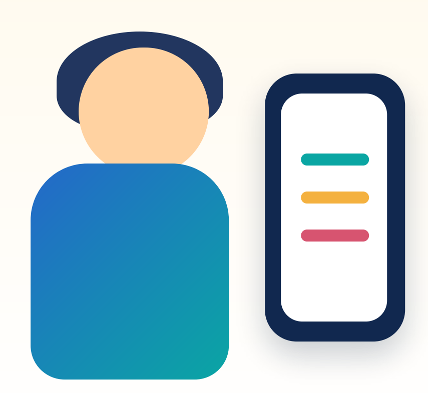
מצב רוח | אנרגיה | שינה | עומס ולחץ | משוב משתמש

## Visual User Journey

**User input → LLM analysis → smart schedule → feedback loop. The schedule adapts to mood, energy, sleep, tasks and real user feedback.**

Central dynamic workflow

### 1 User Opens App



### 2 Daily Check-In

**User input example**
"I slept only 5 hours. I feel stressed and my energy is low today."

- Mood: Stressed
- Energy: 35%
- Sleep: 5h

### 3 Task Input

- Report Writing — Hard
- Team Meeting — Fixed
- Email Review — Easy

### AI Analysis Layer

- Mood Analysis
- Task Classification
- Text → LLM
- Energy Prediction
- Rule-Based Scheduling

### 4 Smart Daily Schedule

- 09:00 — Easy task: Email review
- 11:00 — Fixed task: Team meeting
- 16:00 — Hard task: Report writing

Result: hard work is moved to a later energy window, while fixed tasks stay in place and no overlaps are created.

### 5 Feedback Loop

Low energy today

Schedule changes dynamically

## T Technologies

**React**
Daily check-in, tasks, feedback, and schedule UI.

**Node.js / Express**
REST API for user input, tasks, and generated plans.

**MySQL + REST API**
Stores user history, tasks, schedules, and feedback.

**Python + LLM**
AI services for mood, classification, and prediction.

**Future Development**

Hebrew/Arabic | Multi-user | AI Models | Energy Prediction

React → API → AI

React + Express + MySQL + Python AI

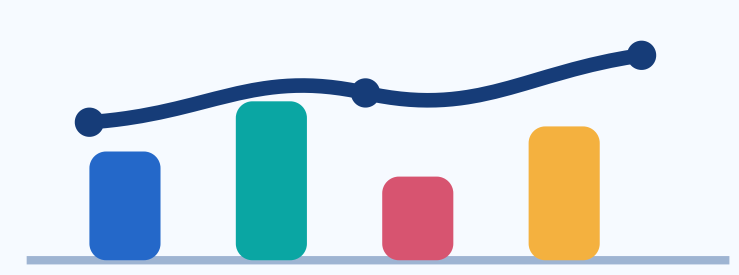
## M Metrics

**A — Accuracy**
Did the AI classify tasks correctly?

**T — Response Time**
Is the schedule generated quickly?

**F — Feedback Quality**
Which feedback improves planning most?

**P — Personalization**
Does planning improve over time?



accuracy | time | feedback | personalization

## AI AI Modules

Text → Mood

**Mood Analysis**
LLM / Sentiment Analysis

Type

**Task Classification**
Easy / Medium / Hard

**Energy Prediction**
Rule-Based + Future ML Models (LSTM)

**Scheduler Engine**
Match Tasks with Energy Levels

**Model Comparison**
Compare LLM models by accuracy, response time, and Hebrew/Arabic support.